Chapter 10 — Electromagnetism

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Q.1 — Multiple Choice Questions (with correct option and reason)

10.1. Current flows north along a power line. The magnetic field over the wire is directed towards:

(a) north **(b)** south
(c) east **(d)** west

Answer: (c) east — Right-hand rule gives east above the wire.

10.2. The radius of the path of a charged particle in a uniform magnetic field is directly proportional to:

(a) the particle's charge **(b)** the particle's momentum
(c) the particle's energy **(d)** the flux density

Answer: (b) the particle's momentum — $r = p/qB$.

10.3. A beam of electrons entering the magnetic field shown is deflected:

(a) into the plane **(b)** out of the plane
(c) towards the bottom of the diagram **(d)** towards the top of the diagram

Answer: (c) towards the bottom of the diagram — Using the left-hand rule for negative charge gives downward deflection for the printed figure.

Figure-dependent; the answer follows the textbook diagram.

10.4. The force on a 1 m wire carrying 1 A perpendicular to the field is called:

(a) magnetic field intensity **(b)** magnetic flux
(c) magnetic induction **(d)** none of these

Answer: (c) magnetic induction — It defines magnetic flux density/induction.

10.5. The unit of flux density is:

(a) $N\ A^{-1}\ m^{-1}$ **(b)** $N\ A\ m^{-1}$
(c) $N\ m\ A^{-2}$ **(d)** $N\ m\ A$

Answer: (a) $N\ A^{-1}\ m^{-1}$ — $B = F/IL$.

10.6. A moving charged particle is surrounded by:

(a) electric field only **(b)** magnetic field only
(c) both electric and magnetic field **(d)** no field

Answer: (c) both electric and magnetic field — A charge has E; motion adds B.

10.7. Magnetic force on charge q moving parallel to B with velocity v is:

(a) $qvB \sin\theta$ **(b)** qvB
(c) zero **(d)** ILB

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Answer: (c) zero — sin 0° = 0.

10.8. The unit N A^{-1} m^{-1} is called:

(a) weber **(b)** tesla
(c) coulomb **(d)** none of these

Answer: (b) tesla — It is the tesla.

10.9. Electrons moving perpendicularly through a uniform magnetic field are:

(a) deflected towards north pole **(b)** deflected towards south pole
(c) deflected along circular path **(d)** not deflected

Answer: (c) deflected along circular path — Magnetic force is centripetal.

10.10. A magnet oscillates in and out of a coil connected to a galvanometer. The galvanometer shows:

(a) constant amplitude left/right deflection **(b)** deflection on one side
(c) no deflection **(d)** left/right deflection with steadily decreasing amplitude

Answer: (d) left/right deflection with steadily decreasing amplitude — Induced current alternates and damps as oscillations die.

Q.2 — Short Answer Questions

10.1. Lenz's law applies to induced currents, not directly to induced emf. Explain.

Lenz's law gives the direction of the induced current, whose magnetic effect opposes the change in flux. The emf direction is inferred from the current it would drive in a closed circuit.

10.2. A square loop moves through a uniform magnetic field with its normal parallel to B. Is an emf induced?

No, if the whole loop remains in the uniform field, the flux linkage does not change, so no emf is induced.

10.3. Does the induced emf always act to decrease the magnetic flux through a circuit?

No. It opposes the change in flux. If flux is increasing, it tries to decrease it; if flux is decreasing, it tries to increase it.

10.4. A magnet is pushed into a solenoid. Why is current produced? What pole appears at the left end?

Changing flux induces an emf and current. By Lenz's law the solenoid's left end becomes the same pole as the approaching magnet, opposing the motion.

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10.5. A bar magnet falls through a fixed metal ring. Will it fall with free-fall acceleration?

No. Induced currents in the ring oppose the motion, producing an upward magnetic force, so acceleration is less than g.

10.6. Which of two charged particles of same mass is deflected most in a magnetic field, fast or slow?

The slower particle, because r = mv/qB is smaller, giving greater curvature.

10.7. An electron and proton enter a magnetic field at right angles with same velocity. Which suffers greater deflection?

The electron, because its much smaller mass gives a much smaller radius.

10.8. Can a single moving proton produce a magnetic field?

Yes. A moving charge is a current and produces a magnetic field.

10.9. A magnetic field is necessary for flux, but a field does not always give flux through a coil. Why?

Flux depends on the perpendicular component: $\Phi = BA \cos \theta$. If the field is parallel to the plane of the coil, flux is zero.

Constructed Response Questions

10.1 A charge lies stationary between opposite poles of two magnets. Is a magnetic force exerted?

No. Magnetic force is $F = qvB \sin \theta$; for $v = 0$ the force is zero.

10.2 When a switch is closed, a metal ring jumps upward. Why? What if battery polarity is reversed?

The increasing current induces a current in the ring that opposes the change, producing repulsion. Reversing polarity still produces a jump because the induced effect opposes the change in flux.

10.3 A coil lies in the x–y plane with B along y. About which axis should it rotate to generate emf?

About the x-axis, so that the flux through the coil changes. Rotation about y would not change the flux.

10.4 Can area and magnetic field both change but no emf be induced?

Yes, if the product BA cos θ, and hence flux linkage, remains constant.

10.5 Does a uniform magnetic field change the kinetic energy of a moving charged particle?

No. The magnetic force is perpendicular to velocity and does no work, so speed and kinetic energy remain unchanged.

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10.6 Uniform E and B act in the same direction. A proton is projected opposite to them with uniform velocity. What happens?

The electric force opposes the motion and slows the proton. The magnetic force is zero because velocity is parallel/antiparallel to B.

10.7 A conductor moving in a magnetic field produces current. Would moving a conductor across a field produce current?

Yes. This is electromagnetic induction: motion changes flux linkage and induces an emf, which drives current in a closed circuit.

10.8 A conducting rod moves on rails and lights a bulb. Where does the electrical energy come from?

From the mechanical work done to move the rod against the magnetic force. Mechanical energy is converted into electrical energy.

10.9 How can a sensitive instrument be saved from stray magnetic fields?

Enclose it in a soft-iron or high-permeability magnetic shield, which diverts magnetic flux around the instrument.

Comprehensive / Long Questions

10.1 Distinguish between magnetic flux and flux density. How are they related?

Flux density B is flux per unit area normal to the field, measured in tesla. Flux $\Phi = BA\cos\theta$ is the total field through an area, measured in weber. Thus $\Phi = B \cdot A$.

10.2 Find an expression for force on a current-carrying conductor in a uniform magnetic field.

For a conductor length L carrying current I at angle θ to B, $F = BIL\sin\theta$. Direction is given by Fleming's left-hand rule.

10.3 State and explain Faraday's law and Lenz's law. Describe factors affecting induced emf.

Faraday: $\varepsilon = -N\Delta\Phi/\Delta t$. Lenz: induced current opposes the change producing it. Induced emf increases with stronger B, faster motion, more turns and larger area.

10.4 Determine the force on a charged particle moving through a uniform magnetic field.

$F = qvB\sin\theta$, perpendicular to v and B. For perpendicular motion it acts as centripetal force, giving circular motion with $r = mv/qB$.

10.5 What is a velocity selector? Explain its working.

Perpendicular E and B fields exert forces qE and qvB in opposite directions. Only particles with $v = E/B$ pass undeflected.

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10.6 Explain how ferrofluids work.

Ferrofluids contain nanometre magnetic particles suspended in liquid. An external field magnetizes them, so the fluid is attracted to magnets and forms spikes along field lines.

Numerical Problems (fully worked)

10.1 A positive charge projected perpendicular at 1500 m s^{-1} experiences force F. At what angle with speed 2000 m s^{-1} is force same?

Given: $v_1 = 1500$, $\theta_1 = 90°$, $v_2 = 2000$

Formula: $qv_1B = qv_2B \sin\theta$

$\sin\theta = 1500/2000 = 0.75$, $\theta \approx 49°$.

Answer: 49°

10.2 Electrons accelerated through 15 kV enter 0.35 T field. Find maximum magnetic force.

Given: V = 15000 V, B = 0.35 T

Formula: $v = \sqrt{(2eV/m)}$, $F = evB$

$v \approx 7.26 \times 10^7$ m s^{-1}.

$F \approx 1.6\times10^{-19} \times 7.26\times10^7 \times 0.35 \approx 4.1 \times 10^{-12}$ N.

Answer: 4.1×10^{-12} N

10.3 Square coil side 15 cm, 60 turns, B = 0.8 T, plane initially perpendicular to field. Turned 30° in 2 s. Find average emf.

Given: N = 60, A = 0.0225 m^2, B = 0.8 T, $\Delta\theta = 30°$, $\Delta t = 2$ s

Formula: $\varepsilon = N B A (1 - \cos30°)/\Delta t$

$\varepsilon \approx 60\times0.8\times0.0225\times0.134/2 \approx 0.07$ V.

Answer: 0.07 V

10.4 A rod in 0.2 T has emf 0.8 V. What field is needed for 2.4 V?

Given: $\varepsilon_1 = 0.8$ V, $B_1 = 0.2$ T, $\varepsilon_2 = 2.4$ V

Formula: $\varepsilon \propto B$

$B_2 = 0.2 \times 2.4/0.8 = 0.6$ T.

Answer: 0.6 T

The book prints 0.67 T; proportionality gives 0.6 T.

10.5 Copper ring radius 4.0 cm, resistance 1.0 mΩ. B increases 0.2 T to 0.4 T in 5 × 10^{-3} s. Find current.

Given: r = 0.04 m, R = 1.0×10^{-3} Ω, $\Delta B = 0.2$ T, $\Delta t = 5\times10^{-3}$ s

Formula: $\varepsilon = A\Delta B/\Delta t$, $I = \varepsilon/R$

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$A = \pi r^2 \approx 5.03\times10^{-3}$ m².

$\varepsilon \approx 0.201$ V, $I \approx 201$ A.

Answer: 201 A

10.6 Coil 10 turns, area 35 cm², perpendicular B = 0.5 T. Pulled out in 1.0 s. Find induced emf.

Given: $N = 10$, $A = 3.5\times10^{-3}$ m², $B = 0.5$ T, $\Delta t = 1$ s

Formula: $\varepsilon = NBA/\Delta t$

$\varepsilon = 10\times0.5\times3.5\times10^{-3} = 1.75 \times 10^{-2}$ V.

Answer: 1.75×10^{-2} V

10.7 Proton accelerated through 6×10^2 V enters perpendicular B = 1.0 T. Find radius.

Given: $V = 600$ V, $B = 1.0$ T, $m = 1.67\times10^{-27}$ kg, $e = 1.6\times10^{-19}$ C

Formula: $r = \sqrt{(2mV/e)}/B$

$r \approx 0.112$ m = 11.2 cm.

Answer: 11.2 cm

10.8 Proton enters B = 0.3 T at 45° with speed 10^6 m s^{-1}. Find radius of circular path.

Given: $v = 10^6$ m s^{-1}, $\theta = 45°$, $B = 0.3$ T

Formula: $r = mv \sin\theta/(eB)$

$r = 1.67\times10^{-27}\times10^6\times0.707/(1.6\times10^{-19}\times0.3) \approx 2.46 \times 10^{-2}$ m.

Answer: 2.46×10^{-2} m

10.9 Three rods length 60 cm move at 2.5 m s^{-1} in different planes; B = 0.5 T along z. Find emf and positive end.

Given: $L = 0.6$ m, $v = 2.5$ m s^{-1}, $B = 0.5$ T

Formula: $\varepsilon = BLv$ for suitable orientation

Rod L_1: $\varepsilon = 0.5\times0.6\times2.5 = 0.75$ V, end a positive.

Rods L_2 and L_3: emf = 0 for the printed orientations.

Answer: L_1: 0.75 V, a positive; L_2: 0; L_3: 0

10.10 Emf 0.5 V induced in 0.4 T. What field gives 1.5 V, other factors same?

Given: $\varepsilon_1 = 0.5$ V, $B_1 = 0.4$ T, $\varepsilon_2 = 1.5$ V

Formula: $\varepsilon \propto B$

$B_2 = 0.4 \times 1.5/0.5 = 1.2$ T.

Answer: 1.2 T

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10.11 A particle passes a velocity selector at 4.96×10^6 m s^{-1}; B = 0.114 T. Find E.

Given: v = 4.96×10^6 m s^{-1}, B = 0.114 T

Formula: E = vB

E = $4.96\times10^6 \times 0.114 \approx 5.65 \times 10^5$ N C^{-1}.

Answer: 5.65×10^5 N C^{-1}

10.12 Conductor PQ length 2 m perpendicular to B = 0.5 T experiences 1 N into the paper. Find current magnitude and direction.

Given: F = 1 N, L = 2 m, B = 0.5 T

Formula: I = F/(BL)

I = 1/(0.5×2) = 1 A.

Direction is Q to P by Fleming's left-hand rule.

Answer: 1 A, Q to P

Textbook checks flagged in this chapter

- MCQ 10.3 is figure-dependent; the answer follows the printed diagram.
- Numerical 10.4: book prints 0.67 T; ε ∝ B gives 0.6 T.

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